

Debrup Dey

Generative AI Engineer

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Professional Summary

Generative AI Engineer with 4 years at Infosys building production-grade agentic and RAG systems — multi-agent orchestration, MCP-driven code-intelligence platforms, and knowledge-graph retrieval. Delivered systems that cut debugging, onboarding, and audit-cycle times by **70–80%**, with **2 patents** filed in agentic AI.

Experience

Specialist Programmer L2 — Infosys STG

Aug 2022 – Present

Code Intelligence Platform

Apr 2025 – Present

(Python, Tree-sitter, Neo4j, Qdrant, MCP, Google ADK, LLMs, FastAPI)

- Architected a platform that turns **50+ repos** into a searchable code knowledge graph — **1M+ relationships** linking functions, files, and services — so developers and AI agents can understand how the code fits together.
- Parsed code with Tree-sitter and indexed it two ways: a Neo4j graph for structure/dependencies and Qdrant embeddings for semantic search.
- Added incremental indexing that re-processes only changed files and their dependents, plus cross-repo linking via API/RPC calls, shared libraries, events, and shared DBs.
- Exposed it through an MCP server (**9 tools**: semantic search, dependency lookup, call-graph, Cypher) any AI agent can consume.
- Built a code-intelligence agent on top — a planner that delegates to specialized sub-agents — that answers questions with citations, speeding debugging, onboarding, and ticket resolution.

Legacy COBOL Comprehension & Knowledge Extraction System

Oct 2024 – Mar 2025

(Python, Custom Parsers, Neo4j, LLMs)

- Developed an AI system to reverse-engineer and document legacy COBOL codebases, accelerating business-logic extraction by **70x** over manual analysis.
- Built a **40k+ node** dependency DAG across **50+ modules**, extracting workflows, business logic, **3600+** glossary terms, and variable lifecycles to recover undocumented tribal knowledge.
- Engineered **division-aware COBOL parsers** with copybook expansion and embedded SQL/CICS extraction to build normalized ASTs.
- Implemented **path-based retrieval** over the graph, enabling deep technical and business-rule search across the modernized codebase.

Agentic Code-RAG System

Apr 2024 – Sept 2024

(Python, LLMs, Agentic RAG, AST Parsing, LangGraph, LlamaIndex, Qdrant)

- Built an **agentic Code-RAG** platform where a router agent dispatches queries across a hybrid retriever (semantic + BM25, with query rewriting and HyDE expansion) and a knowledge-graph pipeline (LLM-generated Cypher traversal), with LLM-based reranking — lifting retrieval recall by **35%** and ranking quality by **28%** (NDCG@10).
- Scaled context-aware coding agents to **1000+ daily queries** across **50+ developers**, cutting task-resolution time by **~70%**.

Multi-Agent Audit Automation System

Mar 2023 – Mar 2024

(Python, LangGraph, LLMs, State Management)

- Built a hierarchical **LangGraph** multi-agent auditing platform: an orchestrator agent infers intent, classifies the audit process, runs a clarification loop, generates a custom audit plan, and routes to **5+ process-specific subagents** (e.g., 3-way match across Invoice/PO/Receipt).
- Implemented **human-in-the-loop** escalation over a shared state store — subagents pause and route approvals through the orchestrator as the single auditor-facing channel, preserving plan, history, and context across hand-offs.
- Automated compliance checks across **10k+ transactions**, cutting audit cycle time by **~80%** at **~95%** match accuracy.

Projects

Manim Animation Agent — AI-Powered Mathematical Animation Generator

- Architected a multi-agent system converting math problems into animated Manim videos via a 4-phase pipeline of **12+ LLM agents** for reasoning, storyboarding, code generation, and vision-based multimodal refinement, with hierarchical orchestration, shared state, validation, and self-correction — cutting animation time from hours to minutes.
- **Tech Stack:** Python, Google ADK, Azure OpenAI, LiteLLM, Manim

Technical Skills

AI/ML: RAG, Agentic RAG, BM25/Hybrid Search, Embeddings, Prompt Engineering, Evaluation/Evals, LlamaIndex, LangChain, Google ADK, LangGraph, MCP, Azure AI, Neural Networks, LLMs

Languages: Python, C++, SQL

Backend: FastAPI, Django

DevOps: Docker, Kubernetes, Git

Databases & Vector Stores: PostgreSQL, Neo4j, Qdrant, ChromaDB

Patents Filed

Belief Space Management for Digital Artifacts

Adaptive Visualization of Digital Artifacts

Awards

- **Winner, Makethon 18** – Developed **Inclusify**, an AI-powered cultural inclusivity platform.
- **3rd Place, IBM watsonx** – Awarded for excellence in generative AI solution development.

Education

Techno Main Salt Lake

B. Tech in Electronics and Communication Engineering

Kolkata, India

2018 – 2022